

Voice Card

direct, simple, practical, plain-spoken, unfiltered, no-buzzwords

Case Study: Content Refresh Priority Queue (Machine Learning) The Problem

Our client manages a site with 30,000 pages. Performance changes constantly, and many pages lose traffic over time. A human editor can only inspect 10 to 20 pages a day. Randomly picking pages wastes time on healthy content, while letting declining pages sit untouched burns organic traffic and revenue.

What I Did & Decided

Instead of forcing editors to guess, I framed this as a Machine Learning scoring and ranking task.

- **Defined a practical target:** Created a needs_review binary flag (1 if traffic trend is down AND 90-day impressions exceed 100). This flagged 13,152 out of 30,000 pages (43.84%) as priority candidates.
- **Avoided hard cutoffs:** A simple if/else rule dumps 13,000+ pages into one unsorted pile. I used ML to output a continuous probability score (0 to 1) to build a smooth priority queue instead.
- **Prevented data leakage:** Explicitly excluded trend_direction from the model's feature set, since it defines the target. The model relies instead on raw signals — page age, days since last update, position, CTR, and impressions.
- **Selected Precision@K as the core metric:** Picked Precision@10/20 over ROC-AUC, because an editor only reviews a small batch each day. The top of the queue has to be accurate.

Outcome

I delivered a complete ML framing blueprint and a runnable notebook pipeline. Instead of digging through a spreadsheet of 30,000 rows by hand, the pipeline gives an editor a clean top-10 focus list to work from — so high-value declining pages get looked at first, not buried in the backlog.

Bio

I'm Jared, a full-stack developer building software, database systems, and practical machine learning pipelines. I focus on turning raw data and complex business problems into clear tools that save time and fix real operational bottlenecks.

Contact & Links

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Looking for someone who can bridge full-stack engineering and machine learning? Let's connect and build something practical.

Before & After: Killing Generic AI Copy

Generic AI Version (Before): "Utilized cutting-edge machine learning methodologies and predictive analytics to seamlessly optimize content performance workflows, driving synergistic traffic recovery and maximizing operational efficiency for editorial teams."

My Edited Version (After): "I turned an unsorted pile of 30,000 web pages into a ranked top-10 daily review list so editors stop wasting time guessing which pages to fix."